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Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

ITEP 414 – System Administration and Maintenance

Week 3 Portfolio Project

Enterprise Server Deployment and Operating System Installation

Program: Bachelor of Science in Information Technology (BSIT)

Course: ITEP 414 – System Administration and Maintenance

Week: 3

Project Type: Individual Portfolio Project

Duration: 5 Hours (2 Hours Lecture + 3 Hours Laboratory)

Submission Platform:

- GitHub Repository
- Learning Management System (LMS)
- LinkedIn

Total Points: 100

Project Title

Enterprise Server Deployment and Operating System Installation

Project Overview

Operating Systems serve as the foundation of every enterprise IT infrastructure. Before a server can host websites, databases, cloud services, or enterprise applications, it must first be properly installed, configured, secured, and documented. In this project, you will assume the role of a **Junior System Administrator** responsible for deploying a new Ubuntu Server for a startup company. You will perform a complete operating system installation, configure essential settings, verify the installation, compare modern boot technologies, and create professional documentation suitable for future administrators.

The deliverables from this project will become part of your professional GitHub portfolio.

Learning Objectives

After completing this project, students should be able to:

Knowledge

Students should be able to:

1. Explain the purpose of an operating system in enterprise environments.
2. Differentiate BIOS and UEFI firmware.
3. Explain the stages of the computer boot process.
4. Compare Ubuntu Server, Windows Server, and Rocky Linux.

Skills

Students should be able to:

- Install Ubuntu Server in a virtual machine.
- Configure server settings during installation.
- Enable secure remote administration using SSH.
- Verify server functionality.
- Document installation procedures.
- Produce professional technical documentation.

Professional Outcomes

By the end of this activity, students will have:

- Installed Ubuntu Server

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- Configured a working Linux server
- Produced a deployment guide
- Created installation documentation
- Updated their GitHub portfolio
- Published a LinkedIn learning reflection

Project Scenario

You have recently joined **ABC Startup Solutions** as a **Junior System Administrator**.

The company plans to deploy its first Linux server that will later be used for:

- File Sharing
- Remote Administration
- Database Hosting
- Web Hosting
- Internal Services

Before deployment, your supervisor instructs you to install Ubuntu Server following company standards and document every step so future administrator can reproduce the installation.

Software Requirements

Install the following before starting:

- Oracle VirtualBox or VMware Workstation
- Ubuntu Server LTS ISO (latest stable version)
- Windows Server Evaluation ISO (for bring-home activity)

Part A – Ubuntu Server Installation (Laboratory)

Objective

Deploy a clean Ubuntu Server virtual machine using recommended enterprise installation practices.

Virtual Machine Specifications

Configure a virtual machine with the following minimum specifications:

Component	Minimum Requirement
Name	Ubuntu-Server-Week03
RAM	4 GB
CPU	2 Virtual Processors
Storage	40 GB (VDI/VMDK)
Network	NAT (or Bridged if instructed)
Optical Drive	Ubuntu Server ISO

Installation Tasks

Follow the installation wizard and configure the following:

1. Language

- Select **English**.

2. Keyboard Layout

- Select the appropriate keyboard layout (e.g., English (US)).

3. Network Configuration

- Accept DHCP unless instructed otherwise.
- Record the assigned IP address.

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4. Hostname

Assign the hostname:
server01

5. User Account

Create a non-root administrative user.

Example:

- Full Name: Your Name
- Username: yourlastname
- Strong Password

6. Disk Partition

Use **Guided – Use Entire Disk**.

Document:

- Partition scheme
- File system used
- Disk size

7. SSH Server

Enable:

- ✓ Install OpenSSH Server

8. Additional Packages

Do not install additional packages unless instructed.

9. Complete Installation

Restart the virtual machine and remove the installation ISO if prompted.

Part B – Server Verification

After installation, verify the server by completing the following tasks.

Task 1 – Login

Log in using the account you created.
Capture a screenshot.

Task 2 – Verify Hostname

Run:

hostname

Capture a screenshot.

Task 3 – Verify IP Address

Run:

ip addr

Capture a screenshot showing the assigned IP address.

Task 4 – Test Internet Connectivity

Run:

ping -c 4 google.com

Capture a screenshot of the successful response.

Task 5 – Update the Server

Run:

sudo apt update

sudo apt upgrade -y

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Capture a screenshot showing the update process.

Task 6 – Verify SSH Service

Run:

`systemctl status ssh`

The service should display **active (running)**.

Capture a screenshot.

Part C – BIOS vs UEFI Comparison

Research and compare BIOS and UEFI.

Prepare a comparison table including:

- Definition
- Boot Process
- Maximum Disk Support
- Partition Style
- Security Features
- Speed
- Advantages
- Disadvantages
- Modern Usage

Write a one-page explanation summarizing why UEFI has largely replaced BIOS in modern computers.

Part D – Boot Process Flowchart

Create a flowchart illustrating the Ubuntu boot process.

Use Draw.io, Lucidchart, Microsoft Visio, or a similar tool.

Your flowchart should include:

1. Power On
2. BIOS/UEFI Initialization
3. Boot Device Detection
4. Boot Loader (GRUB)
5. Linux Kernel
6. init/systemd
7. Services Start
8. Login Prompt

Export the flowchart as PNG and PDF.

Bring-Home Assignment

Install Windows Server Evaluation

Install Windows Server Evaluation in a new virtual machine.

Minimum tasks:

- Install the operating system.
- Assign a computer name.
- Create an administrator password.
- Log in successfully.
- Capture screenshots of the installation.



Comparison Report

Compare:

- Windows Server
- Ubuntu Server
- Rocky Linux

Discuss the following:

- Licensing
- User Interface
- Package Management
- Security
- Performance
- Typical Enterprise Use Cases
- Advantages
- Disadvantages

Present your findings in a comparative table followed by a one-page discussion.

Documentation Requirements

Create a professional **Installation Guide** containing:

- Project Overview
- Objectives
- Software Requirements
- Virtual Machine Specifications
- Step-by-Step Installation Procedure
- Configuration Details
- Verification Commands
- BIOS vs UEFI Comparison
- Boot Process Flowchart
- Windows Server Installation Summary
- OS Comparison Table
- Problems Encountered
- Solutions Implemented
- Lessons Learned
- References

Include screenshots throughout the guide to illustrate each major step.

Mini Project

Professional Installation Manual

Develop a polished installation manual that another system administrator could follow without additional instructions.

Your manual should include:

- Cover Page
- Table of Contents
- Introduction
- Ubuntu Server Installation Guide
- Initial Server Configuration
- Server Verification Procedures
- Troubleshooting Guide
- Best Practices



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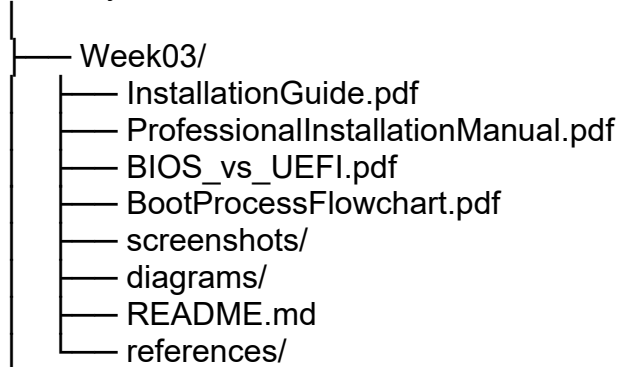
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- References

Export the manual as a PDF.

GitHub Repository Structure

BSIT-SystemAdministration-Portfolio/



README.md Requirements

Your Week03/README.md must contain:

- Project Overview
- Learning Objectives
- Virtual Machine Specifications
- Installation Summary
- Configuration Summary
- Verification Results
- BIOS vs UEFI Highlights
- Embedded Boot Process Flowchart
- Challenges Encountered
- Reflection (300–500 words)
- References

LinkedIn Portfolio Task

Publish a professional LinkedIn post summarizing your Week 3 project.

Include:

- What you installed and configured.
- What you learned about Ubuntu Server deployment.
- Why BIOS/UEFI knowledge matters.
- One screenshot of your virtual machine or boot process flowchart.
- A link to your GitHub repository.
- Relevant hashtags such as **#UbuntuServer #Linux #SystemAdministration #Virtualization #GitHub #BSIT #LearningInPublic**.



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Submission Checklist

Before submitting, verify that your repository contains:

Requirement	Complete
Ubuntu Server Installed	☐
Hostname Configured	☐
Keyboard Configured	☐
Disk Partition Configured	☐
SSH Installed and Running	☐
Login Verified	☐
Internet Connectivity Verified	☐
System Updated	☐
BIOS vs UEFI Comparison	☐
Boot Process Flowchart	☐
Windows Server Evaluation Installed	☐
Windows vs Ubuntu vs Rocky Linux Comparison	☐
Installation Guide	☐
Professional Installation Manual	☐
README.md	☐
GitHub Repository Updated	☐
LinkedIn Reflection Posted	☐

Grading Rubric (100 Points)

Criteria	Excellent (A)	Good (B)	Satisfactory (C)	Needs Improvement (D/F)	Points
Ubuntu Server Installation & Configuration	Ubuntu Server is installed successfully with all required configurations (hostname, keyboard, disk partition, SSH) completed correctly.	Installation is successful with one minor configuration issue.	Installation is functional but several required configurations are incomplete.	Installation failed or multiple critical configurations are missing.	25
Server Verification	All verification tasks (login, hostname, IP, internet, updates, SSH) are completed with complete evidence and correct commands.	Most verification tasks completed with minor omissions.	Some verification tasks completed but evidence is incomplete.	Verification is largely incomplete or incorrect.	15
BIOS vs UEFI Comparison & Boot Process Flowchart	Comparison is technically accurate, comprehensive	Minor inaccuracies or	Basic comparison and flowchart	Incomplete, inaccurate, or missing	15

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	e, and supported by a clear, professional flowchart.	presentation issues.	with limited detail.	comparison/flowchart.	
Windows Server Installation & Operating System Comparison	Windows Server installation is successful, and the comparison with Ubuntu Server and Rocky Linux is accurate, insightful, and well organized.	Minor omissions or superficial comparison.	Installation or comparison lacks depth or completeness.	Missing installation or incomplete comparison.	15
Technical Documentation & Installation Manual	Documentation is complete, professionally formatted, logically organized, and includes clear screenshots, troubleshooting, and references.	Good documentation with minor formatting or content issues.	Documentation is understandable but lacks organization or required sections.	Documentation is incomplete or poorly presented.	20
GitHub Repository & Professional Submission	Repository follows the required structure, includes all deliverables, has a well-written README, and a professional LinkedIn reflection is published.	Minor organizational or submission issues.	Repository is functional but incomplete or lacks polish.	Repository is incomplete, inaccessible, or required submissions are missing.	10
				TOTAL	100 Points

Expected Portfolio Artifact

Upon completing Week 3, you will add a complete **Enterprise Server Deployment** project to your professional portfolio. This artifact demonstrates that you can install and configure a Linux server, validate its readiness for deployment, understand modern boot technologies, compare enterprise operating systems, and produce documentation that another administrator can reliably follow. These are foundational competencies expected of entry-level System Administrators, IT Support Engineers, and Infrastructure Specialists.

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